Computer Networks

Overview of the Internet
Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
  - World Wide Web and HTTP
  - FTP
  - Electronic Mail
  - DNS
- Peer-to-peer paradigm
  - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

BitTorrent: Popular P2P Network

- BitTorrent:
  - Popular peer-to-peer file sharing protocol
- Designed by:
  - Bram Cohen
- Purpose:
  - Sharing large files among many peers
- File sharing in BitTorrent:
  - Collaborative process
- This collaborative process is called:
  - Torrent

Working of BitTorrent

- Large file divided into:
  - Small chunks
- Each peer:
  - Downloads chunks from other peers
  - Uploads chunks to other peers
- Sharing model:
  - Tit-for-tat mechanism
- Result:
  - Faster and distributed downloading
- Peers cooperate:
  - To distribute complete file efficiently

Swarm, Seed, and Leech

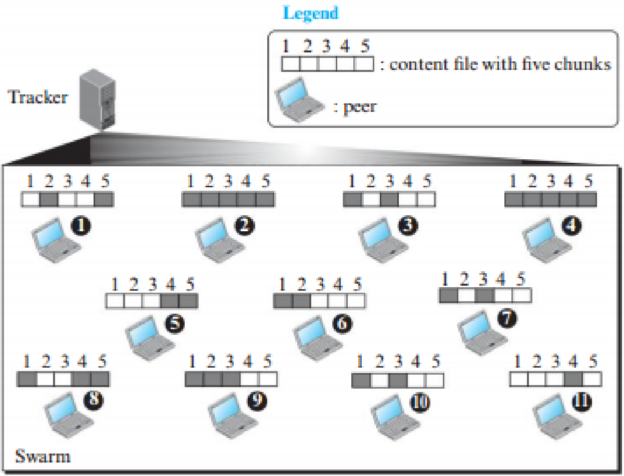
- Group of peers in torrent:
  - Called a swarm
- Seed:
  - Peer with complete file
- Leech:
  - Peer with only part of file
  - Wants remaining chunks
- Swarm contains:
  - Seeds and leeches
- More seeds:
  - Improve download performance

BitTorrent Versions

- Original BitTorrent:
  - Used central node called tracker
- Tracker:
  - Maintains information about peers
- Newer versions:
  - Eliminate tracker
- Modern BitTorrent uses:
  - Distributed Hash Table (DHT)
- DHT provides:
  - Decentralized peer discovery
  - Better scalability

Example of a torrent

Figure 2.57 *Example of a torrent*



Note: Peers 2 and 4 are seeds; others are leeches.

BitTorrent with a Tracker

- Original BitTorrent uses:
  - Central entity called tracker
- Tracker function:
  - Tracks swarm operation
- Torrent contains:
  - Seeds
  - Leeches
  - Tracker
- Content file divided into:
  - Small pieces or chunks
- Different peers may have:
  - Different chunks of file

Joining a Torrent

- New peer wants to download file:
  - Contacts BitTorrent server
- Server provides:
  - Torrent file (metafile)
- Torrent file contains:
  - Information about file pieces
  - Address of tracker
- New peer then:
  - Contacts tracker
- Tracker returns:
  - Addresses of neighboring peers
- New peer now:
  - Joins swarm
  - Starts downloading and uploading chunks

BitTorrent Fairness Policies

- BitTorrent uses policies for:
  - Fair sharing
  - Efficient downloading
  - Preventing overload
- Each peer limits:
  - Number of concurrent neighbors
- Neighbors classified as:
  - Choked or unchoked
  - Interested or uninterested
- Unchoked peers:
  - Currently exchanging data
- Choked peers:
  - Not currently connected

Optimistic Unchoking and Rarest-First

- Every 10 seconds:
  - Peer checks better data-rate neighbors
- Better peers may:
  - Replace slower unchoked peers
- Every 30 seconds:
  - Random peer promoted temporarily
- This process called:
  - Optimistic unchoking
- BitTorrent also uses:
  - Rarest-first strategy
- Rarest-first:
  - Downloads least available pieces first

Trackerless BitTorrent

- Problem with central tracker:
  - Single point of failure
- Newer BitTorrent versions:
  - Remove centralized tracker
- Tracking function distributed using:
  - Distributed Hash Table (DHT)
- Kademlia protocol used for:
  - Finding responsible nodes
- Metadata hash acts as:
  - Key
- Peer list acts as:
  - Value
- Result:
  - Fully decentralized peer discovery